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$$\log_5 x + 4\log_x 5 = 5 \Rightarrow \frac{1}{\log_x 5} + 4\log_x 5 = 5 \overset{u=\log_x 5}{\Rightarrow} \frac{1}{u} + 4u = 5$$

$$\Rightarrow 4u^2 - 5u + 1 = 0 \Rightarrow u = \frac{5 \pm \sqrt{25-16}}{8} = \left\{1, \frac{1}{4}\right\}$$

$$\log_x 5 = 1 \Rightarrow x^1 = 5 \Rightarrow x = 5$$

$$\log_x 5 = \frac{1}{4} \Rightarrow \sqrt[4]{x} = 5 \Rightarrow x = 625$$

References